Deploy

Pipeline status

Loaded: 2026-10-20 01:09

Segmented: 2026-10-20 01:09

Model: K-Means, k=4, silhouette 0.363

Active customers: 5,832

RFM snapshot date is 2011-12-10, the day after the last invoice. 'Active' means net spend > 0 (20 customers who returned everything are excluded).

Clear cache

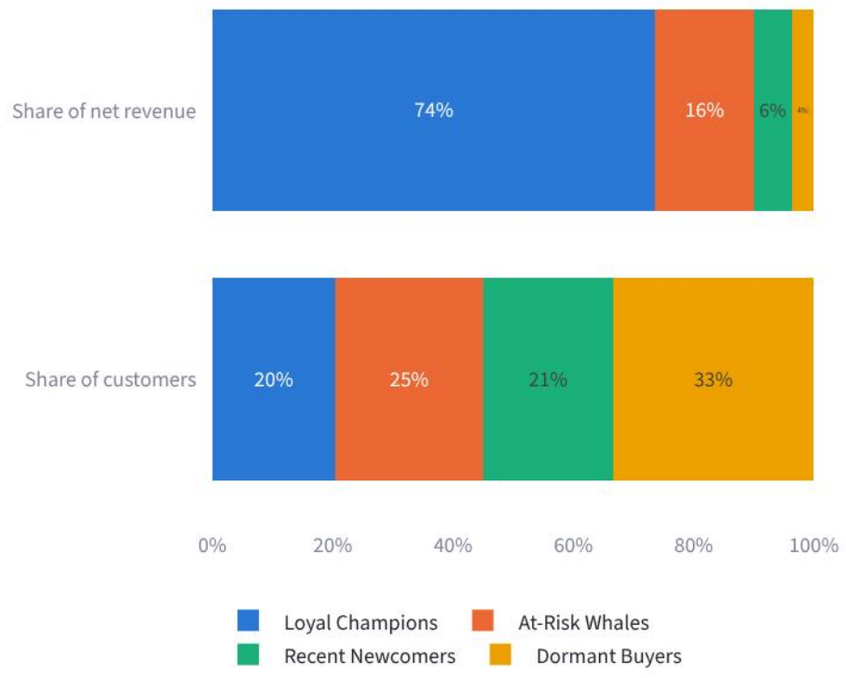
E-Commerce Customer Segmentation & AI Retention Engine

UCI Online Retail II · UK online giftware retailer · Dec 2009 – Dec 2011 · RFM K-Means segments · local CrewAI + Ollama copywriter

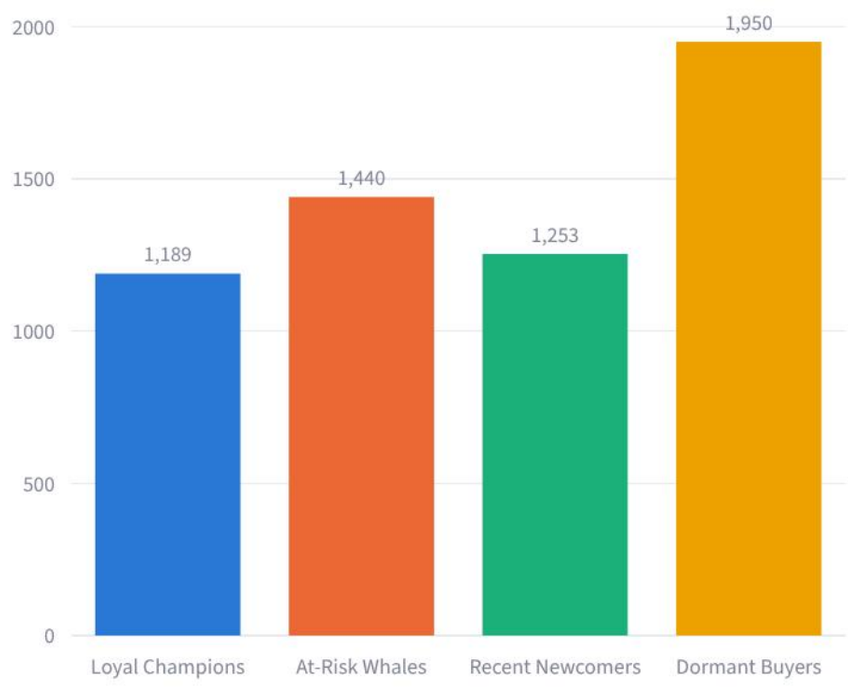
📊 Executive Cohort & LTV | AI Retention Campaigns | ⚙️ Churn Risk Simulator

Active customers	Net revenue (2 yrs)	Net revenue, last 12 full mon…	Avg order value	Churn-risk LTV (3,390 cus…
5,832	£16.36M	£7.93M	£467	£3.29M
		↓ -2.0% vs prior 12		

Revenue concentration: who the revenue comes from

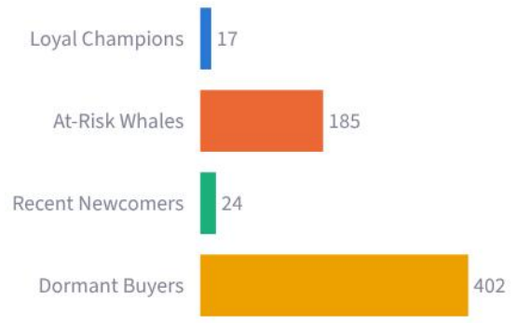


Segment size (customers)

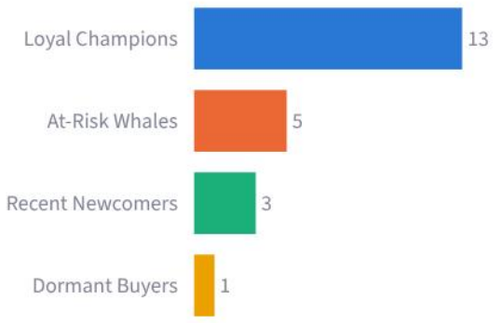


What each segment looks like (medians)

Days since last order



Orders

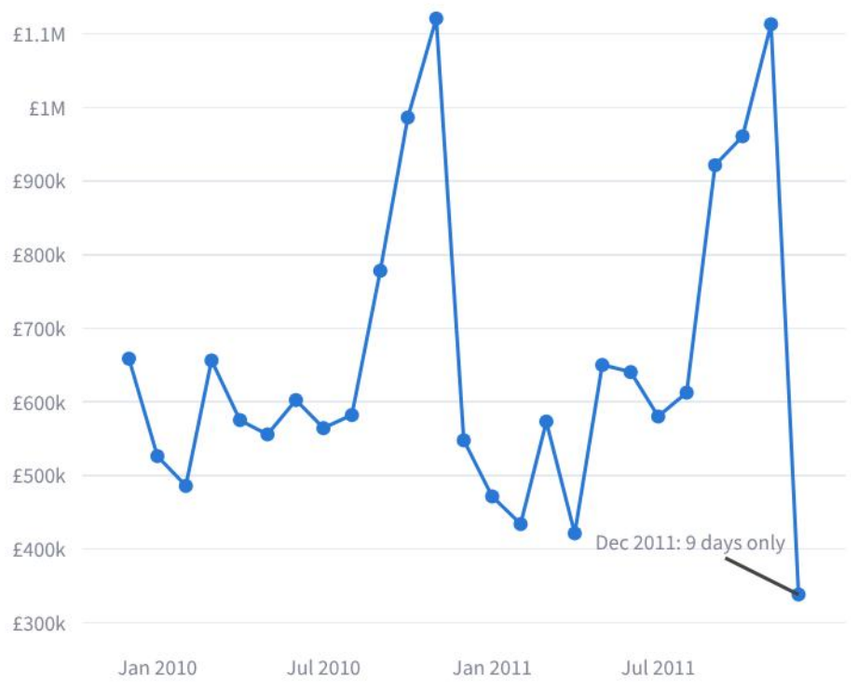


Net spend (£)

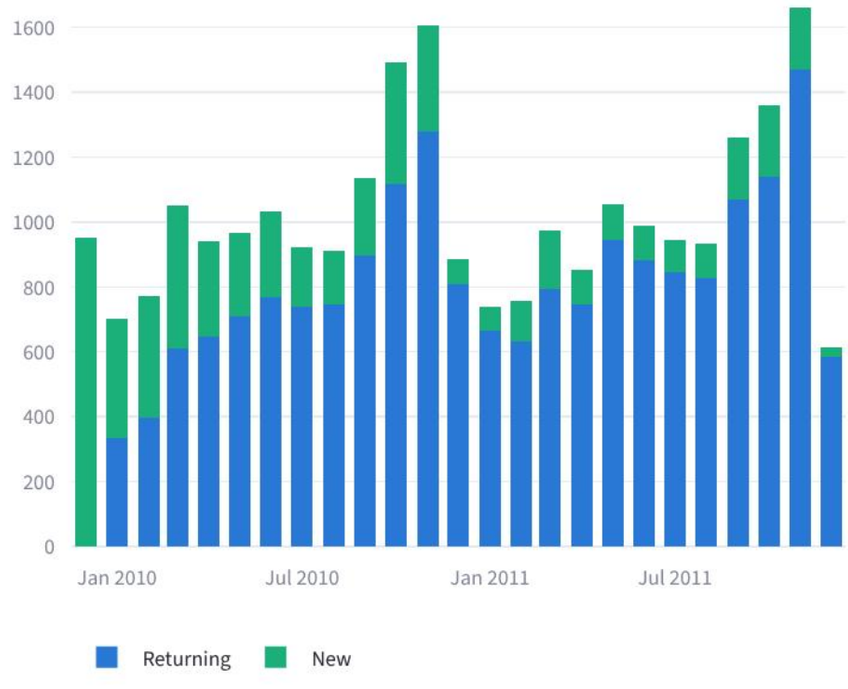


Sales velocity

Net revenue by month (£)



Active customers by month: new vs returning



Dec 2009 counts every customer as 'new' because the data starts there (left-censoring), so new-customer counts are only meaningful from early 2010.

› Segment table and model diagnostics

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Retention campaign generator (local AI)

A CrewAI 'Lifecycle Marketing Analyst' on Ollama `llama3.1` pulls the cohort brief from DuckDB through fixed read-only tools, then drafts the campaign. Nothing leaves this machine. A run takes 1–3 minutes on a laptop.

Segment

At-Risk Whales ⚠ churn risk

Customers (24.7% of base)	Share of net revenue	Median days since last order	Discount cap (policy)
1,440	16.4%	184	15%

Objective: win back lapsing high-value accounts before they churn for good
Preferred incentive: VIP perk (free shipping on the next 3 orders or early access to new ranges) first; a percentage discount only as a secondary hook

Top products	Customers who bought	Revenue
WHITE HANGING HEART T-LIGHT HOLDER	463	£50,052
REGENCY CAKESTAND 3 TIER	404	£34,418
PACK OF 72 RETROSPOT CAKE CASES	321	£5,809
BAKING SET 9 PIECE RETROSPOT	307	£8,135
ASSORTED COLOUR BIRD ORNAMENT	281	£16,845

Ollama model 'llama3.1' isn't pulled (available: ['llama3.2:latest', 'qwen2.5:7b']). Run `ollama pull llama3.1` or set OLLAMA_MODEL to one of the available models.

Generate AI retention campaign

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Churn risk simulator

Lapse threshold: a customer counts as lapsed after this many days without an order

90

Fast-forward: days with no new orders from anyone

90

The threshold re-labels customers by a simple rule. Fast-forward ages every customer's recency and re-scores them with the trained K-Means model (same scaler and centroids), so you can see who drifts into a churn-risk segment if nothing is done.

Lapsed > 90d (51% of base)

2,955

Lapsed customers' LTV

£3.11M

Lapsed Champions (£573.0k LTV)

47

Active vs lapsed by segment at a 90-day threshold



If no one orders for 90 days

Customers in churn-risk segments

5,153

↑ +1,763 vs today

Lifetime value moving into churn risk

£2.85M

Loyal Champions remaining

652

↓ -537

Segment sizes, today vs after

Where customers move